

Contents

The Retarded Nonlocal MOND Kernel Is Unstable on MOND Backgrounds: a Longitudinal Gradient Instability and a Deep-MOND Ghost Close the Nonlocal Door

Abstract	1
1. The door, and what was known	2
2. The candidate, for the record	2
3. Setup and internal checks	3
4. The dispersion	3
5. Mechanism	4
6. Consequences	4
7. Scope, stated honestly	5
8. Attribution	5
9. Two-sided summary	5
10. Reproducibility	5
11. References	6

The Retarded Nonlocal MOND Kernel Is Unstable on MOND Backgrounds: a Longitudinal Gradient Instability and a Deep-MOND Ghost Close the Nonlocal Door

C. P. Zimmerman *Briar Creek Tech · Charlotte, NC · carl@briarcreektch.com*

Version 2026-09-02 · Zenodo [10.5281/zenodo.22253953](https://zenodo.org/record/22253953). Companion to Zenodo [10.5281/zenodo.22242701](https://zenodo.org/record/22242701) (the cluster polytrope) and [10.5281/zenodo.20779562](https://zenodo.org/record/20779562) (the cluster no-go).

Abstract

Purely metric nonlocal realizations of MOND (Deffayet, Esposito-Farèse & Woodard 2011; Woodard 2014; Deffayet & Woodard 2026) carry the MOND force in a retarded nonlocal scalar $X = \square_{\text{ret}}^{-1}(R_{\mu\nu}u^\mu u^\nu)$ through an algebraic function $f(Z)$, $Z = (4c^4/a_0^2)\partial_\mu X\partial^\mu X$, and are the one class that escapes the local no-go for a relativistic MOND completion with $\Phi = \Psi$ lensing and $\alpha_3 = 0$. No perturbation analysis of this kernel on a MOND background exists. We perform it. Around a uniform MOND field ($\bar{Z} = 4y^2$, $y = g/a_0$), in Newtonian gauge and plane symmetry, at leading order in $a_0 \rightarrow 0$ at fixed y , the scalar sector of the in-in theory — the auxiliaries eliminated by their retarded particular solutions — has the dispersion $(e + 3)\omega^4 - (2e - 1)k^2\omega^2 + ek^4 = 0$ at $y = 1$ for the exponential law, with discriminant $1 - 16e < 0$: complex frequencies, $\text{Im}\omega = 0.39ck$. The instability is present for every $y \geq 0.5$ tested ($\text{Im}\omega/ck = 0.21, 0.39, 0.44, 0.36, 0.18, 0.05$ at $y = 0.5, 1, 2, 3, 5, 8$), grows without bound in k (e-folding 1.3×10^3 yr at 1 kpc, 13 yr at 10 pc), is longitudinal only (the transverse sector is stable), is driven entirely by $f''(Z)$ — the μ' term without which there is no MOND — and appears identically for Deffayet–Woodard’s own $f(Z) = \frac{1}{2}Ze^{-\sqrt{Z}/3}$. In the real-frequency deep-MOND window ($y \leq 0.25$) one of the two propagating scalar modes has negative energy. The result holds under both definitions of the theory, the ordinary localized action and the in-in retarded functional, because $\det M = \det D \cdot \det M_{\text{red}}$ and null data removes only the pure-auxiliary poles at $\omega = \pm k$. Every internal check passed first: the GR limit has no scalar metric

mode and the clock mode $\omega^2 = u/(1+u)k^2$ exactly; the static longitudinal response is $1/(\mu_{\parallel}k^2)$ with $\mu_{\parallel} = d(y\mu)/dy = 1 - 2f' - 4\bar{Z}f''$; $\Phi = \Psi$; the localized (X, ξ) kinetic block is $[-4e^{-y}, 1], [1, 0]$, $\det = -1$. The mechanism is the radial gradient instability that killed the khronometric a_{μ} -coupled class: any dynamical carrier of $\mu(|\nabla\Phi|)$ turns μ' into a wrong-sign longitudinal kinetic term in the transition regime. Consequences: the condensate-clock nonlocal candidate we had built the same night, which passed 29 local-Lagrangian gates, is dead at the stability gate; the Deffayet–Woodard kernel inherits the result; and with the local carriers and the aether already closed, every class examined for the strict completion fails. Scope: linear order, WKB background, plane symmetry, this kernel structure; nonlocal form factors on the Weyl or Einstein tensor are not covered and no action for them exists. All numbers are printed by a committed script with checks that can fail and an ablation control.

1. The door, and what was known

A relativistic completion of MOND is asked to deliver, from one action, exact MOND with the exponential law $\mu(y) = 1 - e^{-y}$, no-slip lensing $\Phi = \Psi$, an acceptable full PPN set including the preferred-frame parameters, matter conservation, luminal gravitons, stability, an expanding cosmology and controlled limits (the 13-gate specification of the companion programme). By 2026-09-01 the local carriers were closed by committed computation: elliptic and algebraic constraint carriers by $\alpha_3 = O(1)$ (an instantaneous constraint cannot be retarded), the frame-free scalar by $O(1)$ slip, the clock-plus-curvature carriers by $c_T \neq c$, the acceleration-coupled khronometric carrier by a radial gradient instability on $a_0 < a < 38a_0$, and the aether by $\alpha_1 = -2(K_B + 2)$. One door remained: a genuinely nonlocal, retarded carrier, for which $\alpha_3 = 0$ because the response $1/(k^2 - \omega^2/c^2)$ carries the retardation.

The concrete instance is the Deffayet–Woodard kernel. With a timelike frame $u_{\mu} = \partial_{\mu}\phi$,

$$X = \square_{\text{ret}}^{-1}(R_{\mu\nu}u^{\mu}u^{\nu}), \quad Z = \frac{4c^4}{a_0^2}g^{\mu\nu}\partial_{\mu}X\partial_{\nu}X, \quad \Delta L = \frac{c^4}{16\pi G}a_0^2f(Z)\sqrt{-g},$$

and in the static weak field $X \rightarrow \Psi$, $Z \rightarrow 4|\nabla\Psi|^2c^4/a_0^2 = 4y^2$, so that the potential obeys $\nabla \cdot [(1 - 2f'(Z))\nabla\Psi] = 4\pi G\rho/c^2$: the MOND function is $\mu = 1 - 2f'(4y^2)$. Deffayet & Woodard use $f = \frac{1}{2}Ze^{-\sqrt{Z}/3}$; the exponential law is obtained exactly by $f_{\text{exp}} = 4 - 2(\sqrt{Z} + 2)e^{-\sqrt{Z}/2}$. The kernel is localized by a multiplier, $\xi(\square X - R_{uu})$, whose ordinary-action Hessian in the (X, ξ) pair is indefinite, $[-4e^{-y}, 1], [1, 0]$ with $\det = -1$ (an independent Dirac audit, 2026-09-02): two auxiliary modes, one ghost-signed. The standard resolution is to define the theory in the in-in (Schwinger–Keldysh) sense, with X, ξ as retarded functionals of the metric carrying no Cauchy data. What that theory’s free modes are on a background where the kernel is active had never been computed: Deffayet & Woodard (2026) defer the perturbation analysis; Woodard (2014) lists the ghost question as future work; the certification audit of the kernel in this repository was carried out on Minkowski and FLRW, where $\bar{Z} = 0$ or f is exponentially dead.

We computed it, having first built the strongest candidate the door admits.

2. The candidate, for the record

The condensate-clock nonlocal MOND action (CCNL) replaces Deffayet–Woodard’s mimetic clock and its advected dust by the Aether–Scalar–Tensor condensate clock $K(Q) = -2\Lambda + K_2(Q - Q_0)^2$,

$Q = \sqrt{-(\partial\phi)^2}$, $u_\mu = \partial_\mu\phi/Q$, whose dust is a $\gamma = 2$ polytrope with $c_s^2 = |\Psi|c^2$ inside wells and therefore galaxy-safe (Zenodo 22242701), and uses f_{exp} . It passed 29 committed gates: the galaxy radial-acceleration relation shifts by ≤ 0.001 dex at the full cosmic dust share (Deffayet–Woodard’s pressureless dust shifts it by 0.06–0.24 dex), the off-shell Noether identity of the localized action holds to machine precision with the retarded kernel (so the “hand-imposed retarded operator is not Euler–Lagrange” objection is void: retardation is a choice of solution), $R_{uu}^{(1)}[h^{TT}] = 0$ so $c_T = c$, $\gamma - 1 = -e^{-y}$ with $y = 10^{12}$ at the Cassini radius, the clock’s preferred-frame back-reaction is below 10^{-25} , $\alpha_3 = 0$ by structure, the clock is healthy on its positive branch, and a nonlinear minisuperspace integration keeps the auxiliaries slaved and the constraint conserved to 10^{-10} . None of that survives §4.

3. Setup and internal checks

Newtonian gauge, $ds^2 = -(1+2\Phi)dt^2 + (1-2\Psi)d\mathbf{x}^2$, fields of (t, x) , $c = 1$, $16\pi G = 1$. Background: $\bar{X} = sx$ with $s = ya_0$ (so $\bar{Z} = 4y^2$), $\bar{\xi} = 0$ (a uniform gradient sources no ξ), the clock at rest with charge u_b ($\bar{Q} = Q_0(1+u_b)$, $c_s^2 = u_b/(1+u_b)$). The quadratic action is built from the covariant pieces: $\sqrt{-g}R$ to second order; $\sqrt{-g}a_0^2 f(Z)$ expanded around $\bar{Z}$ keeping every term carrying s ; $\sqrt{-g}\xi(\Box X - R_{uu})$ at second order; $\sqrt{-g}K(Q)$ with an exact-to-second-order expansion of Q . Linear Euler–Lagrange equations, plane waves $e^{i(kx-\omega t)}$, and the 5×5 matrix $M(\omega, k)$ on $(\Phi, \Psi, \delta X, \delta\xi, \delta\phi)$. Leading order in $a_0 \rightarrow 0$ at fixed y removes the vacuum-energy and post-Newtonian terms (all background residuals are $O(a_0^2)$: Λ , $a_0^2 f_0$, $f's^2$, $K_2 Q_0^2 u_b$) and leaves the $O(1)$ MOND structure through $f'(\bar{Z})$, $f''(\bar{Z})$. In-in reduction: the auxiliary rows are eliminated by their particular solutions, the Schur complement $M_{\text{red}} = A - BD^{-1}C$; $\det D = -(k-\omega)^2(k+\omega)^2$ are the pure-auxiliary poles that null data removes.

internal check	result
GR + clock, kernel off	(Φ, Ψ) block $\det = -16k^4$: no scalar metric mode; clock $\omega^2 = \frac{1}{21}k^2 = \frac{u_b}{1+u_b}k^2$ exactly
ordinary local auxiliary block, $y = 1$	$(u_b = \frac{1}{20})$ $[[-4e^{-1}, 1], [1, 0]]$, $\det = -1$: the Dirac audit’s matrix, reproduced independently
static longitudinal response, $y = 1$	$\Psi k^2/\rho = -\frac{1}{4} = -1/(4\mu_{\parallel})$ with $\mu_{\parallel} = 1 - 2f' - 4\bar{Z}f'' = d(y\mu)/dy = 1.000$; $\Phi = \Psi$

4. The dispersion

At $y = 1$ with f_{exp} the reduced (Φ, Ψ) determinant factors to

$$-\frac{256k^4(k^2 - 21\omega^2)[(e+3)\omega^4 - (2e-1)k^2\omega^2 + ek^4]e^{-1}}{105(k-\omega)^2(k+\omega)^2},$$

the clock factor times a quartic whose discriminant is $(2e-1)^2 - 4e(e+3) = 1 - 16e < 0$. Its roots are $\omega^2/k^2 = 0.388 \pm 0.570i$, i.e. $\omega/k = 0.734 \pm 0.388i$: a growing and a decaying pair, growth $\propto k$.

test	outcome
y -scan, f_{exp}	real ω^2 at $y = 0.10$ (0.056, 0.897) and 0.25 (0.174, 0.717); complex at $y = 0.5, 1, 2, 3, 5, 8$ with $\text{Im } \omega/k = 0.21, 0.39, 0.44, 0.36, 0.18, 0.05$
Deffayet–Woodard’s $f = \frac{1}{2}Ze^{-\sqrt{Z}/3}$	complex at $y = 0.5, 1, 2, 4$ ($\text{Im } \omega/k$ up to 0.48); real at 0.25 and 8
ablation $f'' \rightarrow 0$ at fixed f'	stable at $y = 1, 2$ (one real mode, $\omega^2/k^2 = 0.30, 0.61$): the instability is the f'' term
ablation $f' \rightarrow 0$ at fixed f''	unstable ($\text{Im } \omega/k = 0.60, 0.52$)
transverse propagation, $\mathbf{k} \perp \nabla \bar{X}$	stable at $y = 0.5, 1, 2, 4$ ($\omega^2/k^2 = 0.14, 0.30, 0.61, 0.93$): longitudinal only
energy sign, deep-MOND window	$y = 0.10$: $\omega/k = 0.236$ ($E > 0$) and 0.947 ($E = -7.8 \times 10^3$); $y = 0.25$: 0.417 (+), 0.847 (-4.1×10^2)
metric content of the unstable mode, $y = 1$	$ \Phi ^2 + \Psi ^2$ share 6.4%, (X, ξ) share 86%: nonzero, hence a mode of the reduced metric equation
e-folding times, $y = 1$	1 kpc: 1.3×10^3 yr; 10 pc: 13 yr; 1000 AU: 2.4 days
e-folding times, $y = 8$	1 kpc: 10^4 yr; 10 pc: 100 yr; 1000 AU: 18 days

Because $\det M = \det D \cdot \det M_{\text{red}}$, the quartic’s roots are modes of the ordinary local action as well; the in-in prescription removes only $\det D = 0$, the pure-auxiliary modes at $\omega = \pm k$. The unstable modes carry metric content and survive under either definition.

5. Mechanism

MOND requires $\mu' \neq 0$. The static AQUAL operator is elliptic and safe, with longitudinal coefficient $\mu_{\parallel} = \mu + y\mu' > 0$ for the exponential law, and the kernel reproduces exactly that static response (§3). Its time-dependent completion is not safe: the quadratic kernel Lagrangian contributes $-4f_1(\partial_t \delta X)^2 + 4(f_1 + 2\bar{Z}f_2)(\partial_x \delta X)^2$, a wrong-sign time-kinetic term for X in the localized pair, and the f_2 (curvature, μ') part of the longitudinal gradient term is what turns the coupled (Φ, Ψ, X, ξ) dispersion complex once $\bar{Z}$ is finite. With $f'' = 0$ — f linear in Z , μ constant, no MOND — the sector is stable. This is the mechanism that killed the khronometric a_{μ} -coupled class on $a_0 < a < 38a_0$ (2026-08-31): a dynamical carrier of $\mu(|\nabla \Phi|)$ converts μ' into a longitudinal instability in the transition regime. Retardation does not cure it; it only relocates the carrier.

6. Consequences

1. **CCNL is dead at the stability gate**, its 29 other passes notwithstanding.
2. **The Deffayet–Woodard kernel inherits the result.** Their 2026 model retains the same X , Z , and an algebraic f ; the difference in the clock sector does not enter at this order (the clock decouples from the kernel at linear order around a vacuum background, $\bar{R}_{\mu\nu} = 0$).
3. **The nonlocal door is closed at linear-WKB order.** With the local constraint carriers (α_3), the frame-free scalar (slip), the curvature-coupled clock (c_T), the acceleration-coupled

clock (radial instability) and the aether (α_1) closed by committed computation, every class examined for the strict completion fails. In the companion programme’s language this is Outcome B, a no-go, not Outcome A.

4. **What survives is unchanged:** MOND phenomenology with the acceleration scale $a_0 = c^2 \sqrt{\Lambda/32\pi}$ as input, the falsifiable $a_0 \propto \sqrt{\rho_{\text{DE}}(z)}$, and the condensate dark field of the AeST class, whose cluster yield is 23–33% of the core residual (Zenodo 22242701). No relativistic completion in hand passes the stability gate.

7. Scope, stated honestly

Linear order. A uniform-gradient WKB background, $a_0 \rightarrow 0$ at fixed y ; since the growth is $\propto k$, the short-wavelength regime where WKB is best is the dangerous one. Plane symmetry and Newtonian gauge. The kernel structure: u -projected $R_{\mu\nu}$, algebraic $f(Z)$, no higher-derivative terms; Deffayet & Woodard state their philosophy is an IR modification, and a UV completion with higher spatial derivatives capping the growth at some k_{max} is not part of any published action. Not covered: nonlocal form factors acting on the Weyl or Einstein tensor (a “field-dependent spin-2” completion, for which no action exists in the literature or in this repository); the fully nonlinear problem; bounded backgrounds with their own scale. The negative-energy mode in deep MOND is a linear result on the same background.

8. Attribution

The kernel is Deffayet, Esposito-Farèse & Woodard (2011), Woodard (2014) and Deffayet & Woodard (2026). The Dirac structure of the localized pair for this candidate is the independent 2026-09-02 audit in the repository (it also caught a placeholder check in the candidate’s gate script, since replaced by a derived one). The radial-gradient mechanism for μ' -carrying relativistic MOND is the repository’s khronometric result of 2026-08-31. The perturbation analysis of the retarded kernel on a MOND background, the in-in reduction with its internal checks, the f'' ablation and the transverse contrast are new here as far as searched; no such analysis is known to us in the literature.

9. Two-sided summary

For the nonlocal programme: the retarded kernel genuinely does what the local carriers cannot — $\alpha_3 = 0$, $\Phi = \Psi$ at 1PN, exponentially screened PPN, luminal gravitons, conservation as a Noether identity of the localized action, and, with a condensate clock, galaxy-safe dust. Those are real and were computed.

Against: on any background where the kernel is active, the scalar sector is unstable in the transition regime with growth times of years at parsec scales and unbounded in k , and ghostly in deep MOND; the instability is the MOND function’s own curvature μ' and cannot be removed without removing MOND; Deffayet–Woodard’s own interpolation function shows it. The last door is shut at this order.

10. Reproducibility

- In the directory `qwen_claude_field_theory/ closure_2026/ candidate_ccnl_2026/: ccnl_inin_linear_scalar_2026.py` (12 checks, `rc=0`; output `.out`): §3–5. The physics checks are findings that can fail to materialize; the $f'' \rightarrow 0$ ablation is the control.

- `ccnl_mond_gates_2026.py` (29 checks, `rc=0`; `MUTATE=1` breaks the galaxy comparison): §2.
- `ccnl_action_dirac_audit_2026.py` with its tests and manifest (the independent audit): the (X, ξ) block.
- `CCNL_MOND_CANDIDATE.md`, `NONLOCAL_KERNEL_LINEAR_INSTABILITY_VERDICT.md`: the gate table and the verdict.

Quarantine: a_0 (both footings, 9.36×10^{-11} and $1.13 \times 10^{-10} \text{ m s}^{-2}$), κ and the clock's charge are inputs. Nothing here derives them.

11. References

- Deffayet C., Esposito-Farèse G., Woodard R. P., 2011, PRD 84, 124054 (arXiv:1106.4984)
- Deffayet C., Esposito-Farèse G., Woodard R. P., 2014, PRD 90, 064038 (arXiv:1405.0393)
- Deffayet C., Woodard R. P., 2026, JCAP 04, 081 (arXiv:2512.10513)
- Deser S., Woodard R. P., 2013, JCAP 11, 036 (arXiv:1307.6639)
- Skordis C., Złośnik T., 2021, PRL 127, 161302 (arXiv:2007.00082)
- Woodard R. P., 2014, Can. J. Phys. 93, 242 (arXiv:1403.6763)
- Zimmerman C. P., 2026, Zenodo 10.5281/zenodo.20779562 (the cluster no-go); 10.5281/zenodo.22242701 (the cluster polytrope)